

Section: Particles and Waves

Topic: Wave-Particle Duality

This question explores the photoelectric effect and the energy of photons.

(a)

Calculate the energy of a photon

Use:

$$E = hf$$

Where:

- $h = 6.63 \times 10^{-34} \text{ J s}$
- $f = 7.5 \times 10^{14} \text{ Hz}$

$$E = 6.63 \times 10^{-34} \times 7.5 \times 10^{14} = 4.97 \times 10^{-19} \text{ J}$$

(b)

Calculate the maximum kinetic energy of the emitted electrons

Use the photoelectric equation:

$$E_{\text{photon}} = \text{Work function} + E_{\text{kinetic}} \quad \Rightarrow \quad E_{\text{kinetic}} = E_{\text{photon}} - \phi$$

Given:

- $\phi = 2.7 \times 10^{-19} \text{ J}$
- $E_{\text{photon}} = 4.97 \times 10^{-19} \text{ J}$

$$E_{\text{kinetic}} = 4.97 \times 10^{-19} - 2.7 \times 10^{-19} = 2.27 \times 10^{-19} \text{ J}$$

 Revision Tips

- The energy of a photon is proportional to frequency:

$$E = hf$$

- The **work function** is the minimum energy needed to eject an electron from a material
 - If the photon's energy is **less than** the work function → **no electrons emitted**
 - Maximum kinetic energy = photon energy – work function
 - All energies must be in **joules**, unless stated otherwise
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